README — STAR to Count Data

Collects all ReadsPerGene.out.tab files produced by STAR for DSMZ RNA-seq samples.

Extracts the unstranded count column (column 2) and the Ensembl gene ID with version (column 1).

Merges all samples into a single count table (genes × samples).

Maps Ensembl IDs (version removed) to gene_name using a GENCODE GTF.

Saves two RDS files:
• DSMZ_star2count.rds — merged counts with gene_id_ver (Ensembl with version) + sample columns.
• DSMZ_count_gene.rds — tidy table with Ensembl_ID_with_version, Ensembl_ID (no version), gene_name, then sample columns.

Inputs

STAR outputs: one ReadsPerGene.out.tab per sample in:
/home/chu25/data/star/count_data

The script assumes STAR's default column order:
1. Ensembl gene ID with version
2. Unstranded counts
3. Strand 1 counts
4. Strand 2 counts
(We use column 2.)

GENCODE GTF for gene symbol mapping:
/home/chu25/data/GTF/Homo_sapiens.GRCh38.107.gtf

Outputs

Written to:
/home/chu25/data/dsmz

DSMZ_star2count.rds — merged raw counts; first column gene_id_ver, remaining columns are sample counts.

DSMZ_count_gene.rds — merged counts with three annotation columns:
• Ensembl_ID_with_version (former gene_id_ver)
• Ensembl_ID (version stripped, e.g., ENSG00000141510)
• gene_name (from GTF)
followed by one column per sample.

How samples are named

Sample names are derived from filenames by removing the exact suffix:
*.ReadsPerGene.out.tab

Example:
NG-13640_BC3_lib236396_5774_5.ReadsPerGene.out.tab
→ NG-13640_BC3_lib236396_5774_5

Notes & assumptions

• Rows starting with N_ in STAR outputs are dropped (they are STAR summary lines, not genes).
• Missing genes in some samples become 0 after the full join.
• If you need stranded counts, change the extracted column from 2 to 3 or 4 accordingly.
• This script does not run STAR; it only merges its outputs.